Simulating a Distributed Cassandra Cluster Using Docker (Multi-Node Setup)

1. Install Prerequisites

Ensure the following are installed:

- **Docker**
- **Docker Compose (optional but recommended)**

Check installation:

docker --version
docker compose version

<u>Using Docker Compose</u>

This method creates **3 Cassandra nodes forming a cluster**.

Step 1: Create Project Directory

mkdir cassandra-cluster
cd cassandra-cluster

```
ubuntu@Suhel-PC:~$ mkdir cassandra-cluster
ubuntu@Suhel-PC:~$ cd cassandra-cluster/
```

Step 2: Create docker-compose.yml

Create a file:

nano docker-compose.yml

Add the following configuration:

```
version: "3"

services:

  cassandra1:
    image: cassandra:4.1
    container_name: cassandra1
    environment:
      - CASSANDRA_CLUSTER_NAME=DemoCluster
      - CASSANDRA_SEEDS=cassandra1
    ports:
      - "9042:9042"
    networks:
      - cassandra-net

  cassandra2:
    image: cassandra:4.1
    container_name: cassandra2
```

```
    environment:
      - CASSANDRA_CLUSTER_NAME=DemoCluster
      - CASSANDRA_SEEDS=cassandra1
    depends_on:
      - cassandra1
    networks:
      - cassandra-net

  cassandra3:
    image: cassandra:4.1
    container_name: cassandra3
    environment:
      - CASSANDRA_CLUSTER_NAME=DemoCluster
      - CASSANDRA_SEEDS=cassandra1
    depends_on:
      - cassandra1
    networks:
      - cassandra-net

networks:
  cassandra-net:
```

Step 3: Start Cassandra Cluster

Run:

docker compose up -d

```
  cassandra-net:
ubuntu@Suhel-PC:~/cassandra-cluster$ docker compose up -d
WARN[0000] /home/ubuntu/cassandra-cluster/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid pot
ential confusion
WARN[0000] No services to build
[+] up 4/4
 ✔Network cassandra-cluster_cassandra-net Created                0.1s
 ✔Container cassandra1                    Created                0.2s
 ✔Container cassandra2                    Created                0.1s
 ✔Container cassandra3                    Created                0.1s
```

Check running containers:

docker ps

```
ubuntu@Suhel-PC:~/cassandra-cluster$ docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED          STATUS          PORTS                                                  NAMES
9657129fc67f   cassandra:4.1   "docker-entrypoint.s…"   8 seconds ago    Up 7 seconds    7000-7001/tcp, 7199/tcp, 9042/tcp, 9160/tcp            cassandra2
6e94cae085fc   cassandra:4.1   "docker-entrypoint.s…"   8 seconds ago    Up 7 seconds    7000-7001/tcp, 7199/tcp, 9042/tcp, 9160/tcp            cassandra3
7ea6bd32d58e   cassandra:4.1   "docker-entrypoint.s…"   9 seconds ago    Up 7 seconds    0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp            cassandra1
73a4f0f8b6c4   cassandra:4.1   "docker-entrypoint.s…"   56 minutes ago   Up 56 minutes   7000-7001/tcp, 7199/tcp, 9042/tcp, 9160/tcp            reverent_torv
alds
```

Step 4: Verify Cluster Nodes

Enter first container:

docker exec -it cassandra1 bash

```
ubuntu@Suhel-PC:~/cassandra-cluster$ docker exec -it cassandra1 bash
root@7ea6bd32d58e:/#
```

Run:

nodetool status

```
root@7ea6bd32d58e:/# nodetool status
Datacenter: datacenter1
=======================
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
--  Address     Load       Tokens  Owns (effective)  Host ID                               Rack
UJ  172.20.0.3  80.68 KiB  16      ?                 6d2d432d-61ce-425a-8d5e-11ae455831eb  rack1
UN  172.20.0.2  80.75 KiB  16      100.0%            5e1b6c66-690c-4b04-99f2-256a47903085  rack1

root@7ea6bd32d58e:/#
```

Meaning:

- **U** = Up
- **N** = Normal

All **3 nodes are part of the distributed cluster**.

Step 5: Connect Using CQLSH

Inside container:

cqlsh

```
ubuntu@Suhel-PC:~/cassandra-cluster$ docker exec -it cassandra1 bash
root@7ea6bd32d58e:/# cqlsh
Connected to DemoCluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.10 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh>
```

Step 6: Create Keyspace With Replication

CREATE KEYSPACE university
WITH replication = {
'class':'SimpleStrategy',
'replication_factor':3
};

Explanation:

- Replication factor **3**
- Data copies stored on **3 nodes**

```
cqlsh> CREATE KEYSPACE university
   ... WITH replication = {
   ... 'class':'SimpleStrategy',
   ... 'replication_factor':3
   ... };

cqlsh>
```

Step 7: Create Table

USE university;

```
cqlsh> USE university;
cqlsh:university>
```

CREATE TABLE students (
id int PRIMARY KEY,
name text,
department text
);

```
cqlsh> USE university;
cqlsh:university> CREATE TABLE students (
                ... id int PRIMARY KEY,
                ... name text,
                ... department text
                ... );



cqlsh:university>
```

Step 8: Insert Data

INSERT INTO students (id,name,department)
VALUES (1,'Rahul','CS');

INSERT INTO students (id,name,department)
VALUES (2,'Anita','IT');

```
cqlsh:university> INSERT INTO students (id,name,department)
                ... VALUES (1,'Rahul','CS');
cqlsh:university>
cqlsh:university> INSERT INTO students (id,name,department)
                ... VALUES (2,'Anita','IT');
cqlsh:university>
```

Step 9: Query Data

SELECT * FROM students;

```
cqlsh:university> SELECT * FROM students;

 id | department | name
----+------------+-------
  1 |         CS | Rahul
  2 |         IT | Anita
```

Step 10: Check Data Distribution

Run:

nodetool ring

This shows how data is **distributed across nodes using consistent hashing**.

```
root@7ea6bd32d58e:/# nodetool ring

Datacenter: datacenter1
==========
Address         Rack        Status State   Load            Owns                Token
                                                                               8916297008348656128
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -8730618394699765205
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -8090178835458312249
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -7472197803556112171
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -7018678219118279436
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -6407856813783048774
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -6000826814375852420
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -5183154881763501626
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -4587852512803102375
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -4107399000537922086
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -3371483307827028445
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -2710788183290116346
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -2292110500379426788
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -1541833260111101966
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -1048590494698834814
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   -492709459781910878
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             -81226251620299316
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   4432037130096045 40
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   858841372581244420
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             1508778300893599321
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   2070521719253225565
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   2467844571601938356
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             3127801596590642618
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   3726265287547381444 4
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             4105947049106756694
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   4760565934222785811
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   5186852435503915201
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             5884581349930343822
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   6537054262078593084
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             6980623250165282111
172.20.0.3      rack1       Up     Joining 80.68 KiB       ?                   7594430779007551301
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             8071920360857494370
172.20.0.2      rack1       Up     Normal  80.75 KiB       100.00%             8916297008348656128

  Warning: "nodetool ring" is used to output all the tokens of a node.
  To view status related info of a node use "nodetool status" instead.
```

Step 11: Stop Cluster

docker compose down